

electron/git-blame.cjs

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Overview

Adds a new synchronous Git blame helper module at `electron/git-blame.cjs`. The file introduces three exported functions: `parseBlamePorcelain`, `blameFileAtRef`, and `blameFileOwnership`. It also enables strict mode and a `Map` cache.

Key changes

- `parseBlamePorcelain(stdout)` (added lines 8-34) parses `git blame --line-porcelain` output into `lineAuthors` and a sorted `owners` array.
- `blameFileAtRef(repoRoot, ref, relPath)` (added lines 37-76) runs `git blame` via `spawnSync`, caches results in `blameCache`, and returns `{ lineAuthors, owners, ref, relPath, error }`.
- `blameFileOwnership(root, relPath)` (added lines 79-81) is a convenience wrapper for `HEAD`.
- The module exports the three functions (line 83) and uses `require("node:child_process")` (line 3).

Impact

- Provides a synchronous API for blame lookups; the call blocks the event loop.
- Caching reduces repeated Git calls for the same `(root, ref, relPath)` key.
- Errors are returned as a trimmed string (up to 300 chars).
- The module is isolated; no existing code paths are modified.

Risks & follow-ups

- **Blocking:** `blameFileAtRef` uses `spawnSync`; ensure it is not invoked on UI threads.
- **Security:** `relPath` is trimmed but not escaped; validate against command-injection.
- **Cache growth:** monitor memory usage; consider eviction if many unique keys accumulate.
- **Testing:** add unit tests for `parseBlamePorcelain` and `blameFileAtRef` to cover empty output, errors, and typical use cases.